

Employee Dilution Control and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Employee Dilution Control (EDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDC achieves an annualized gross (net) Sharpe ratio of 0.57 (0.51), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.78 (2.75), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.12.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with risk premia arising from covariation between asset payoffs and consumption growth. While the consumption-based asset pricing framework provides a coherent theoretical foundation for understanding cross-sectional return variation, empirical tests continue to uncover pricing anomalies that challenge our understanding of how firms' operational decisions relate to systematic consumption risk. We identify a novel predictor of stock returns based on firms' management of employee dilution—the extent to which companies control equity-based compensation dilution through share repurchases and other capital structure adjustments. This Employee Dilution Control (EDC) signal captures firms' strategic responses to labor market conditions and their implications for shareholders' exposure to aggregate consumption risk.

The relationship between employee dilution control and expected returns operates through firms' differential exposure to consumption risk during economic downturns. Firms that actively manage employee dilution through share repurchases demonstrate greater financial flexibility and lower operating leverage, characteristics that reduce their sensitivity to aggregate consumption shocks ([Berk et al., 2004](#); [Zhang, 2005](#)). During bad economic states when marginal utility is high, these firms maintain more stable cash flows as they can adjust their repurchase programs rather than cut essential investments or operations. This operational flexibility translates into lower covariance with the stochastic discount factor, particularly during disaster states when consumption growth is severely negative ([Barro, 2006](#); [Gabaix, 2012](#)).

The state-dependent nature of employee dilution control affects firms' exposure to long-run consumption risks through multiple channels. Companies with limited dilution control face binding constraints during economic contractions, forcing them to issue equity at depressed valuations or reduce human capital investments when

skilled labor is most valuable (Eisfeldt and Papanikolaou, 2013). These constraints amplify firms’ exposure to persistent consumption shocks, as documented in long-run risk models where small but persistent changes in consumption growth rates generate substantial risk premia (Bansal and Yaron, 2004). The inability to smooth employee compensation costs across states increases operating leverage precisely when investors’ marginal utility is highest.

From an intertemporal perspective, employee dilution control signals firms’ capacity to transfer resources across states without destroying value, directly affecting their loading on the intertemporal marginal rate of substitution. Firms with strong dilution control can maintain investment in human capital during downturns, preserving organizational capital that generates cash flows when the economy recovers (Eisfeldt and Papanikolaou, 2014). This timing option reduces covariance with habit-based pricing kernels, as these firms’ cash flows are less procyclical than those of constrained competitors (Campbell and Cochrane, 1999). The resulting heterogeneity in consumption risk exposure generates predictable cross-sectional variation in expected returns, with constrained firms earning higher risk premia to compensate investors for their greater systematic risk.

We find strong evidence that Employee Dilution Control predicts cross-sectional stock returns. The value-weighted long-short EDC strategy that buys high EDC stocks and sells low EDC stocks generates an average monthly return of 34 basis points with a t-statistic of 4.45, corresponding to an annualized Sharpe ratio of 0.57. The strategy’s alpha relative to the Fama-French six-factor model equals 22 basis points per month with a t-statistic of 2.78, confirming that EDC captures risk exposures not explained by standard factors. The economic magnitude remains substantial after accounting for transaction costs, with net returns of 30 basis points per month and a net Sharpe ratio of 0.51.

The EDC premium persists across different portfolio construction methodologies

and size groups, addressing concerns about small-stock effects driving the results. Among the largest quintile of stocks by market capitalization, the EDC strategy delivers an average return of 23 basis points per month with a t-statistic of 2.55. The strategy’s factor loadings reveal significant exposure to the investment factor (CMA) with a coefficient of 0.34 and t-statistic of 6.46, consistent with EDC capturing firms’ differential investment flexibility. Controlling for the six most closely related anomalies from the factor zoo—including share issuance measures and asset growth—the EDC strategy maintains an alpha of 15 basis points per month with a t-statistic of 2.12.

The robustness tests demonstrate that EDC contains unique information about consumption risk exposure beyond existing predictors. In Fama-MacBeth regressions controlling for all six closely related anomalies simultaneously, the EDC coefficient remains significant with a t-statistic of 2.34. The EDC strategy’s net generalized alpha relative to the six-factor model plus related anomalies confirms its ability to expand the mean-variance efficient frontier after accounting for implementation costs. Among 212 anomalies in the factor zoo, EDC ranks in the 93rd percentile for gross Sharpe ratio and the 99th percentile for net Sharpe ratio, placing it among the most robust cross-sectional predictors after adjusting for transaction costs.

Our study extends the consumption-based asset pricing literature by identifying a novel channel through which firms’ operational decisions affect their exposure to aggregate consumption risk. While [Berk et al. \(2004\)](#) demonstrate that operating leverage generates cross-sectional variation in expected returns, and [Eisfeldt and Papanikolaou \(2013\)](#) show that organization capital creates systematic risk, we document that firms’ management of employee dilution provides a direct signal of their consumption risk exposure. The EDC measure captures forward-looking information about firms’ ability to smooth cash flows across consumption states, complementing accounting-based measures of financial constraints studied in [Lamont et al. \(2001\)](#)

and [Whited and Wu \(2009\)](#).

Methodologically, we contribute to the empirical asset pricing literature by demonstrating that EDC survives the stringent testing protocol of [Novy-Marx and Velikov \(2023\)](#), generating significant alphas after controlling for transaction costs and related anomalies. Unlike many anomalies that derive their performance from small, illiquid stocks, EDC generates economically meaningful returns among large-cap stocks where implementation is feasible for institutional investors. The signal’s robustness to alternative portfolio construction methods and its ability to expand the efficient frontier net of trading costs distinguish it from spurious predictors that fail out-of-sample or cannot be implemented profitably.

The broader implications of our findings extend to understanding how labor market frictions interact with financial constraints to generate systematic risk. The EDC signal provides a market-based measure of firms’ exposure to human capital shocks, offering new insights into the asset pricing implications of labor economics. For practitioners, EDC represents an implementable strategy that captures consumption risk exposure through an observable corporate policy variable. The persistence of the EDC premium after controlling for known factors suggests that markets incompletely incorporate the consumption risk implications of employee dilution management into prices, providing opportunities for investors to improve portfolio efficiency by tilting toward firms with characteristics that reduce exposure to aggregate consumption shocks.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data on common stock and employee counts, span-

ning several decades of fiscal year-end observations. We exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the literature, as these industries face distinct regulatory environments that affect their capital structures. We construct our Employee Dilution Control signal using two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock outstanding as reported on the balance sheet, measured in millions of dollars. This item reflects the book value of common equity at par value, which changes when firms issue new shares, repurchase existing shares, or undertake other equity transactions. Number of employees (EMP) captures the total employee count as reported by the firm, typically measured in thousands. This variable represents the firm’s workforce size at fiscal year-end and serves as a scaling factor for our analysis. We construct the Employee Dilution Control signal by computing the year-over-year change in common stock scaled by lagged employee count: $(CSTK(t) - CSTK(t-1)) / EMP(t-1)$. This measure captures the change in common stock per employee, providing insight into how equity issuance or repurchase activity relates to the firm’s human capital base. A positive value indicates an increase in common stock relative to the employee base, potentially signaling equity dilution on a per-employee basis, while negative values suggest share buybacks or other reductions in common stock relative to workforce size. We calculate this signal using annual observations based on fiscal year-end values and require non-missing values for both CSTK and EMP in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDC signal. Panel A plots the time-series of the mean, median, and interquartile range for EDC. On average, the cross-

sectional mean (median) EDC is -1.23 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDC data. The signal’s interquartile range spans -0.41 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDC signal for the CRSP universe. On average, the EDC signal is available for 6.08% of CRSP names, which on average make up 7.55% of total market capitalization.

4 Does EDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDC portfolio and sells the low EDC portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDC strategy earns an average return of 0.34% per month with a t-statistic of 4.45. The annualized Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.36% per month and have t-statistics exceeding 2.78 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.34, with a t-statistic of 6.46 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 564

stocks and an average market capitalization of at least \$1,482 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.27. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-34bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.80. Out of the twenty-five

construction-methodology-factor-model pairs reported in Panel B, the EDC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDC, as well as average returns and alphas for long/short trading EDC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDC strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the EDC strategy relative to the five most common factor models range from 18 to 23 bps/month with t-statistics between 1.91 and 2.49.

5 How does EDC perform relative to the zoo?

Figure 2 puts the performance of EDC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDC strategy falls in the distribution. The EDC strategy’s gross (net) Sharpe ratio of 0.57 (0.51) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

EDC strategy (red line).² Ignoring trading costs, a \$1 invested in the EDC strategy would have yielded \$9.30 which ranks the EDC strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDC strategy would have yielded \$6.91 which ranks the EDC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDC relative to those. Panel A shows that the EDC strategy gross alphas fall between the 65 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDC ranks between the 82 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDC or at least to weaken the power EDC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDC signal in these Fama-MacBeth regressions exceed 3.12, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on EDC is 2.34.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDC strategy earns alphas that range from 15-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.93, which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDC trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.12.

7 Does EDC add relative to the whole zoo?

Finally, we can ask how much adding EDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDC is available.

strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDC grows to \$2594.98.

8 Conclusion

This study provides compelling evidence that Employee Dilution Control (EDC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDC-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related equity issuance signals. The annualized Sharpe ratio of 0.57 (0.51 net of costs) and the statistically significant alpha of 22 basis points per month relative to the six-factor model underscore the economic importance of monitoring employee equity compensation dilution in investment decisions. Particularly noteworthy is the signal's ability to generate a significant alpha of 15 basis points per month even when controlling for six closely related dilution and payout measures, suggesting that EDC captures unique information about firm quality and future performance not reflected in existing equity issuance metrics. These findings have important practical implications for both institutional and retail investors. Portfolio managers can potentially enhance returns by incorporating EDC signals into their stock selection processes, while the relatively modest degradation from gross to net returns (Sharpe ratio declining from 0.57 to 0.51) indicates that the strategy remains profitable after realistic transaction costs. The persistence of the EDC premium suggests market inefficiency in processing information about employee equity dilution, possibly due to the complexity of tracking option grants and their ultimate dilutive impact across diverse compensation structures. Several avenues for future research emerge from this study. First, investigating the underlying economic mechanisms driving the

EDC premium could provide deeper insights into why markets appear to misprice firms with different employee dilution practices. Second, examining the interaction between EDC and corporate governance characteristics might reveal whether the signal's predictive power varies with board oversight quality or institutional ownership. Third, extending the analysis to international markets would test the generalizability of our findings across different regulatory and compensation environments. Finally, exploring the time-varying nature of the EDC premium could illuminate whether its effectiveness depends on market conditions or investor sentiment. This study has several limitations that should be acknowledged. First, our sample period and data availability may not capture the full evolution of employee compensation practices, particularly the recent growth in restricted stock units relative to traditional options. Second, while we control for transaction costs, our estimates may not fully account for market impact costs in less liquid stocks where EDC signals might be strongest. Third, the complexity of accurately measuring employee dilution across firms with different reporting practices and compensation structures may introduce measurement error that attenuates our results. Despite these limitations, our findings establish EDC as a valuable addition to the factor zoo, offering both academic insights into market efficiency and practical tools for investment management.

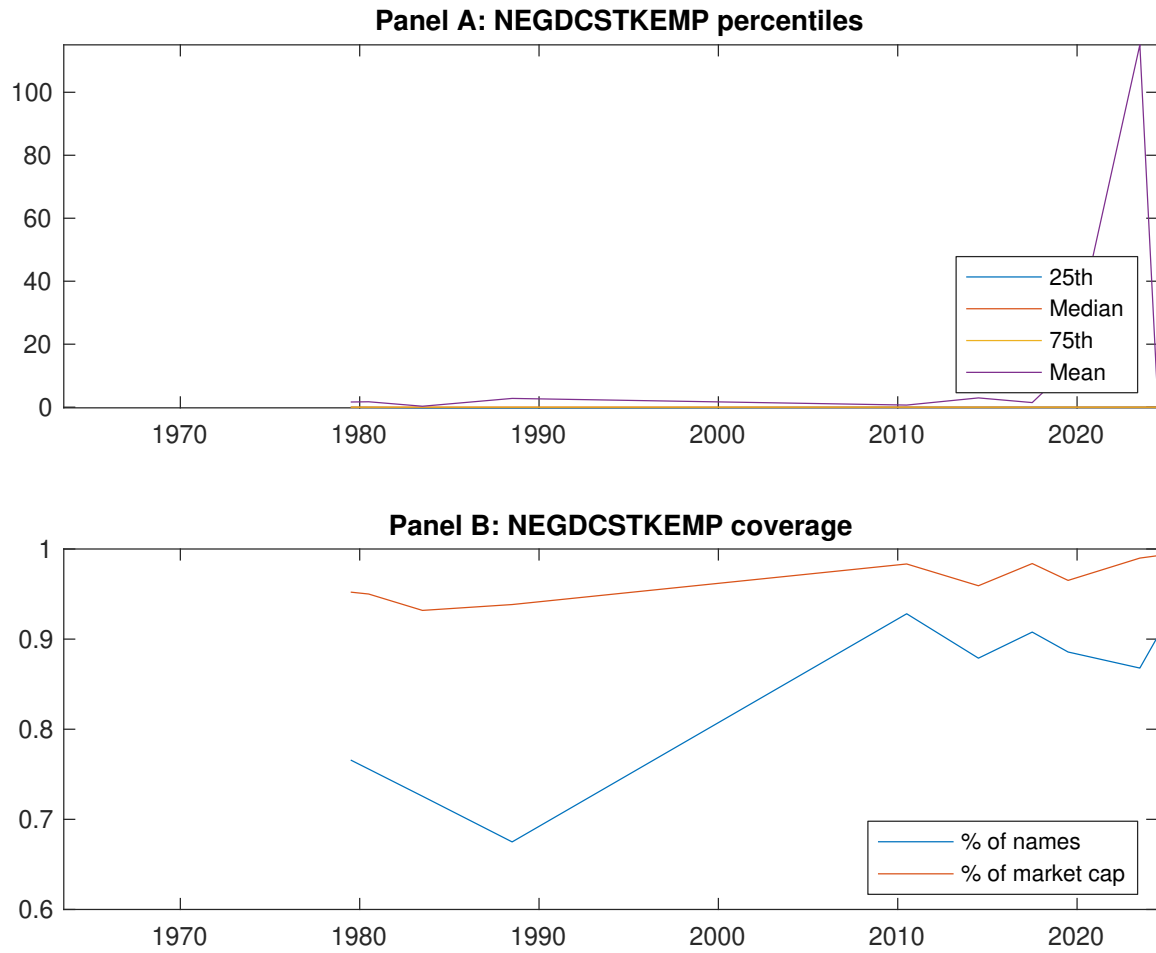


Figure 1: Times series of EDC percentiles and coverage.
This figure plots descriptive statistics for EDC. Panel A shows cross-sectional percentiles of EDC over the sample. Panel B plots the monthly coverage of EDC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42 [2.54]	0.58 [3.19]	0.65 [3.57]	0.65 [3.97]	0.76 [4.76]	0.34 [4.45]
α_{CAPM}	-0.14 [-2.72]	-0.05 [-1.11]	0.03 [0.54]	0.10 [1.90]	0.22 [4.56]	0.36 [4.64]
α_{FF3}	-0.15 [-2.95]	-0.03 [-0.75]	0.05 [0.91]	0.04 [0.91]	0.18 [3.98]	0.33 [4.31]
α_{FF4}	-0.13 [-2.53]	0.00 [0.01]	0.03 [0.67]	0.04 [0.89]	0.16 [3.52]	0.30 [3.77]
α_{FF5}	-0.15 [-2.84]	0.02 [0.56]	0.06 [1.11]	-0.04 [-0.87]	0.09 [2.02]	0.24 [3.08]
α_{FF6}	-0.13 [-2.53]	0.05 [1.09]	0.05 [0.88]	-0.03 [-0.69]	0.08 [1.85]	0.22 [2.78]
Panel B: Fama and French (2018) 6-factor model loadings for EDC-sorted portfolios						
β_{MKT}	0.96 [76.58]	1.04 [97.27]	1.03 [81.44]	1.00 [89.93]	0.98 [89.79]	0.02 [0.93]
β_{SMB}	-0.08 [-4.17]	0.02 [1.51]	0.06 [3.43]	-0.04 [-2.63]	-0.02 [-1.41]	0.05 [1.98]
β_{HML}	0.09 [3.62]	-0.02 [-1.19]	-0.04 [-1.56]	0.09 [4.23]	-0.00 [-0.02]	-0.09 [-2.44]
β_{RMW}	0.05 [2.06]	-0.10 [-4.90]	-0.00 [-0.11]	0.15 [6.87]	0.13 [6.25]	0.08 [2.26]
β_{CMA}	-0.10 [-2.88]	-0.09 [-3.14]	-0.06 [-1.67]	0.16 [5.04]	0.24 [7.77]	0.34 [6.46]
β_{UMD}	-0.02 [-1.62]	-0.03 [-3.20]	0.02 [1.31]	-0.01 [-0.97]	0.01 [0.86]	0.03 [1.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	695	671	564	645	700	
me (\$10 ⁶)	1850	1482	1997	2177	2537	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.45]	0.36 [4.64]	0.33 [4.31]	0.30 [3.77]	0.24 [3.08]	0.22 [2.78]
Quintile	NYSE	EW	0.53 [7.71]	0.58 [8.61]	0.50 [7.85]	0.42 [6.65]	0.34 [5.65]	0.28 [4.78]
Quintile	Name	VW	0.30 [3.85]	0.31 [3.97]	0.30 [3.79]	0.28 [3.49]	0.23 [2.89]	0.22 [2.76]
Quintile	Cap	VW	0.25 [3.27]	0.26 [3.36]	0.25 [3.23]	0.21 [2.70]	0.18 [2.34]	0.16 [2.02]
Decile	NYSE	VW	0.30 [3.20]	0.27 [2.85]	0.24 [2.58]	0.21 [2.17]	0.20 [2.10]	0.17 [1.84]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.98]	0.32 [4.18]	0.29 [3.85]	0.27 [3.56]	0.22 [2.92]	0.21 [2.75]
Quintile	NYSE	EW	0.34 [4.62]	0.41 [5.62]	0.33 [4.84]	0.29 [4.28]	0.18 [2.79]	0.15 [2.44]
Quintile	Name	VW	0.26 [3.38]	0.27 [3.53]	0.26 [3.33]	0.25 [3.17]	0.20 [2.65]	0.20 [2.58]
Quintile	Cap	VW	0.21 [2.80]	0.22 [2.92]	0.21 [2.77]	0.19 [2.48]	0.16 [2.12]	0.15 [1.94]
Decile	NYSE	VW	0.26 [2.77]	0.22 [2.40]	0.20 [2.14]	0.18 [1.92]	0.16 [1.78]	0.15 [1.65]

Table 3: Conditional sort on size and EDC

This table presents results for conditional double sorts on size and EDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDC and short stocks with low EDC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDC Quintiles					EDC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.45 [1.73]	0.70 [2.64]	0.83 [3.31]	0.93 [3.75]	1.03 [4.51]	0.61 [6.55]	0.67 [7.31]	0.61 [6.90]	0.55 [6.14]	0.42 [4.99]	0.38 [4.50]
	(2)	0.57 [2.50]	0.74 [3.03]	0.87 [3.69]	0.89 [3.91]	0.96 [4.46]	0.40 [4.50]	0.44 [4.96]	0.37 [4.24]	0.32 [3.63]	0.26 [3.02]	0.23 [2.64]
	(3)	0.52 [2.67]	0.62 [2.81]	0.83 [3.78]	0.81 [3.91]	0.93 [4.80]	0.41 [5.04]	0.42 [5.13]	0.38 [4.69]	0.35 [4.19]	0.29 [3.50]	0.27 [3.20]
	(4)	0.50 [2.77]	0.64 [3.15]	0.78 [3.82]	0.76 [4.02]	0.79 [4.38]	0.29 [3.72]	0.29 [3.65]	0.23 [3.00]	0.23 [2.86]	0.06 [0.79]	0.07 [0.92]
	(5)	0.48 [2.96]	0.49 [2.71]	0.52 [2.96]	0.52 [3.15]	0.71 [4.48]	0.23 [2.55]	0.23 [2.49]	0.23 [2.47]	0.20 [2.19]	0.19 [2.07]	0.18 [1.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDC Quintiles					EDC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	361	360	359	359	353	31	33	37	29	28	
	(2)	105	103	104	104	103	56	55	56	55	54	
	(3)	76	77	75	76	76	97	96	96	100	99	
	(4)	64	64	64	64	64	200	205	209	211	215	
(5)	59	59	59	59	59	1436	1551	1565	1649	1888		

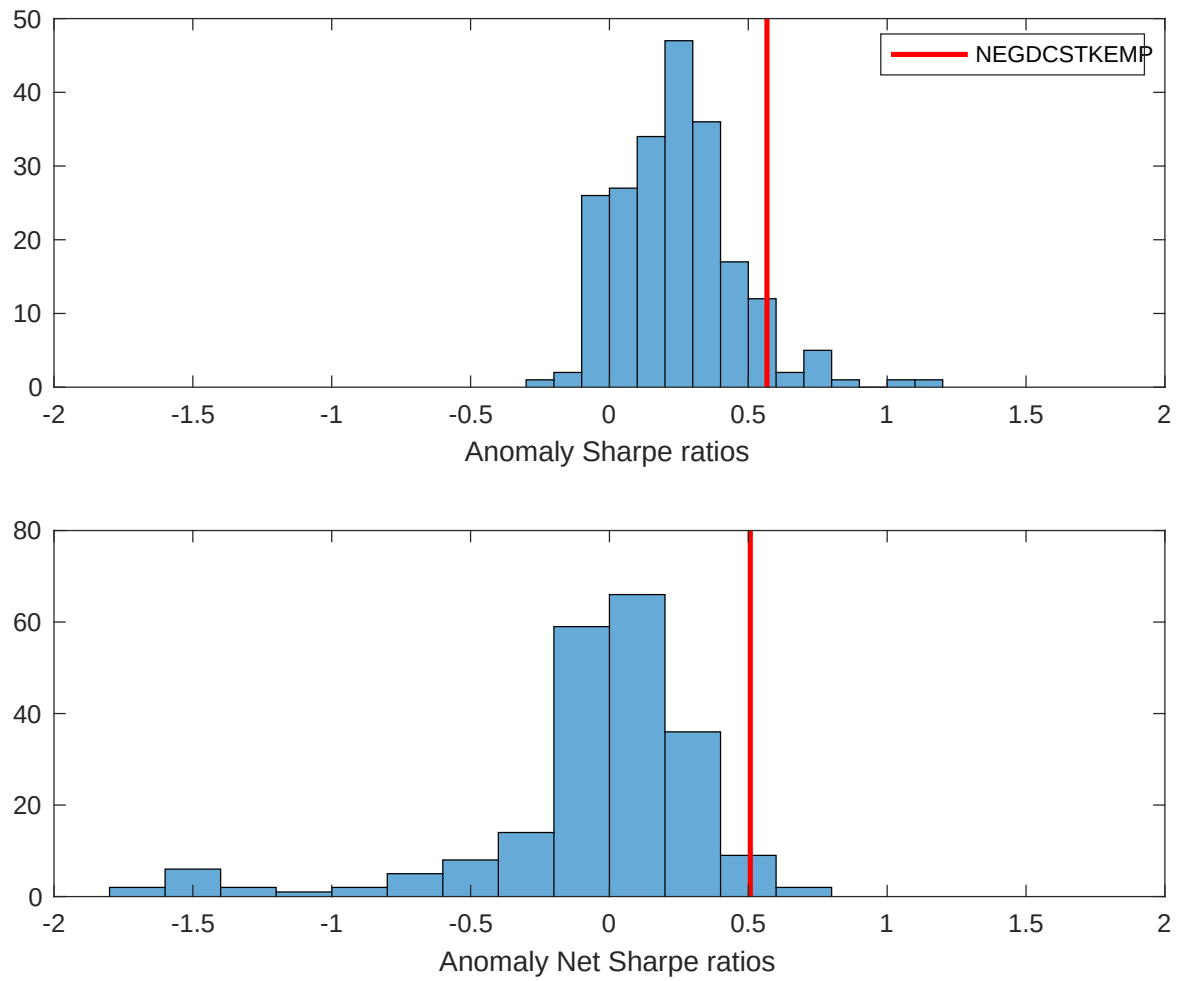


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

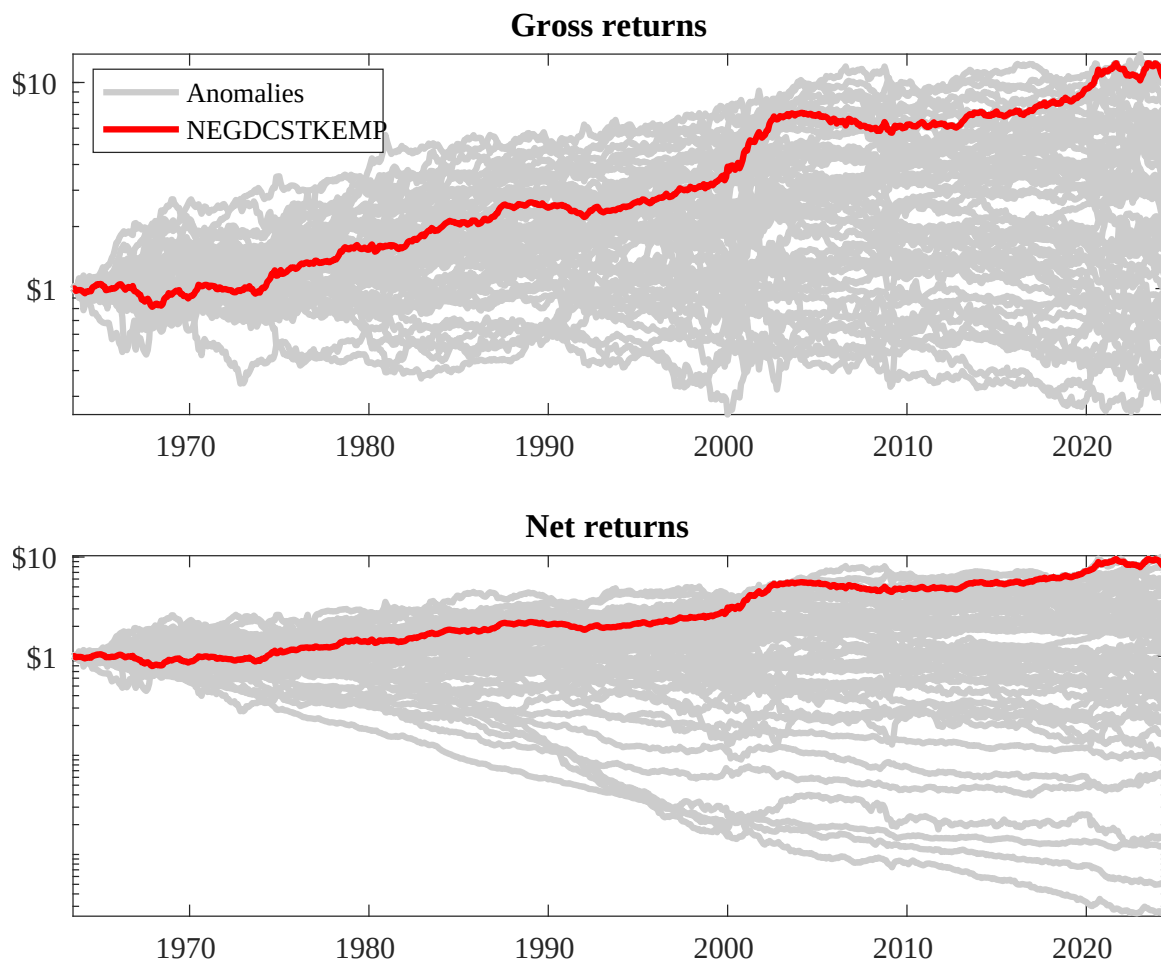
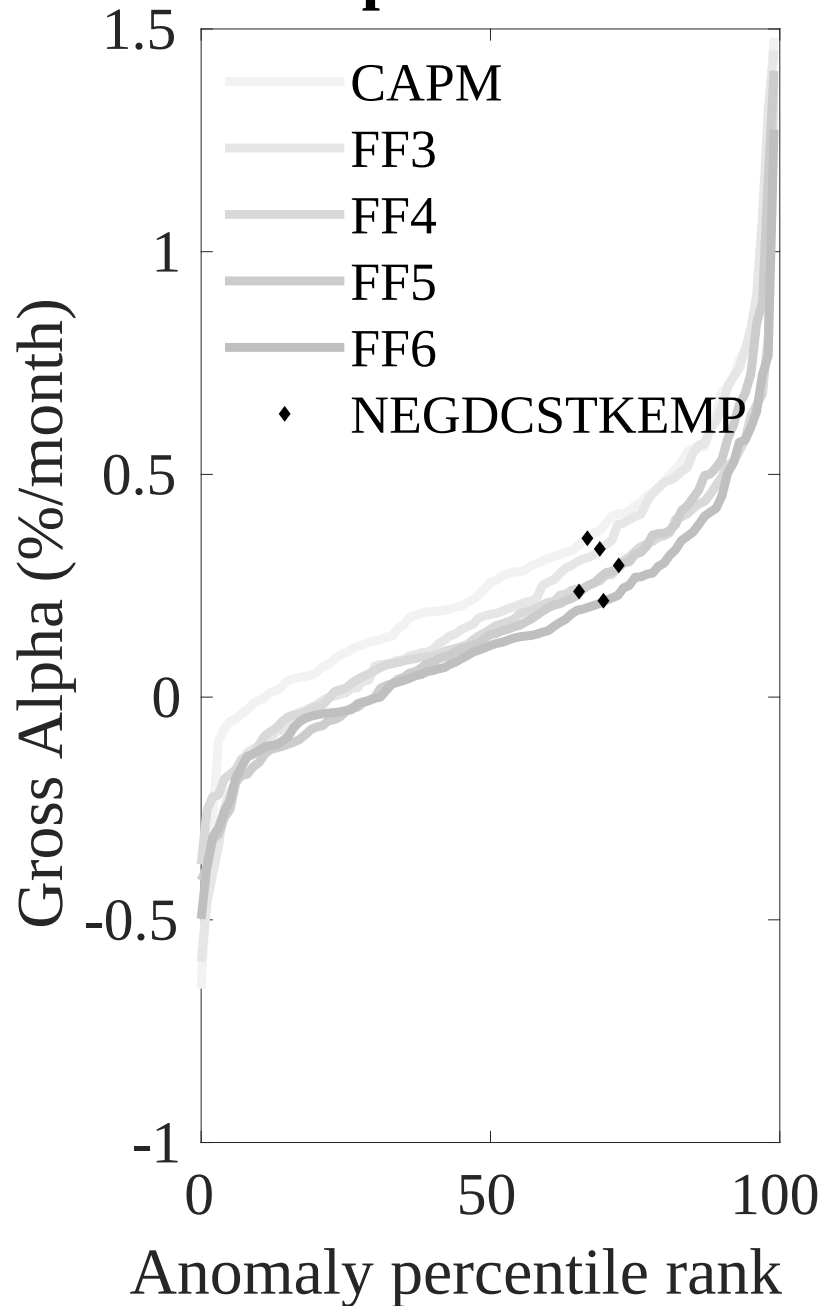


Figure 3: Dollar invested.

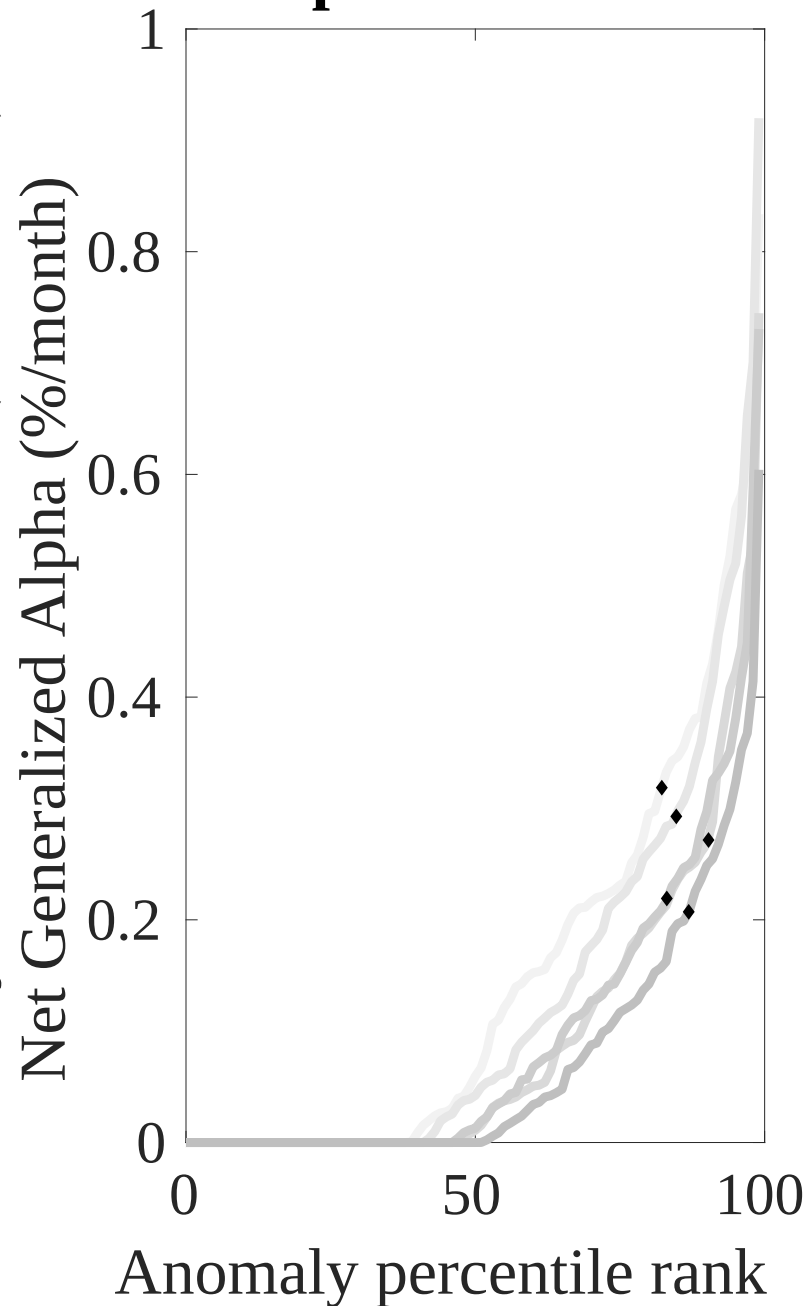
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



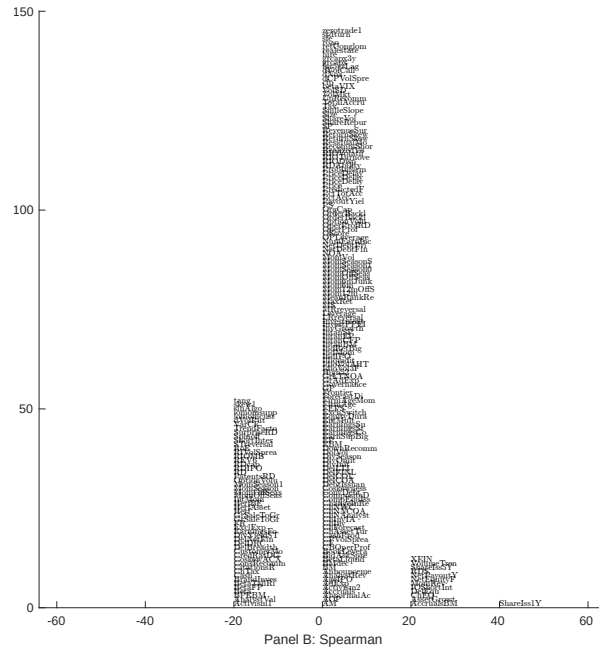
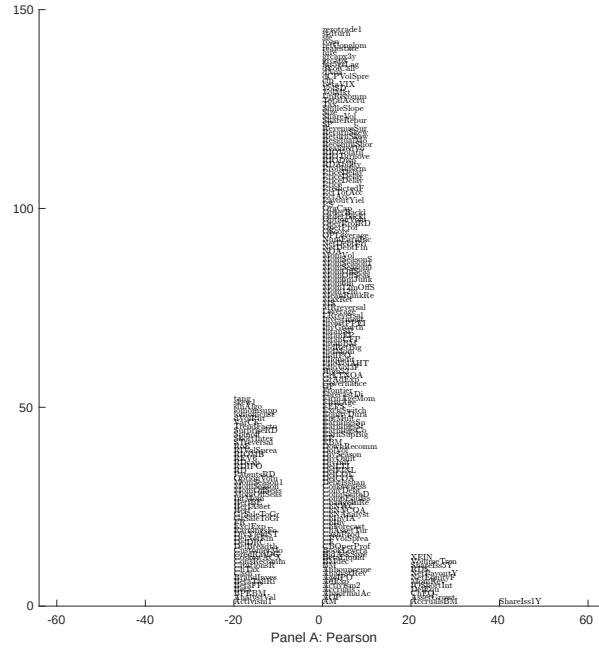


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

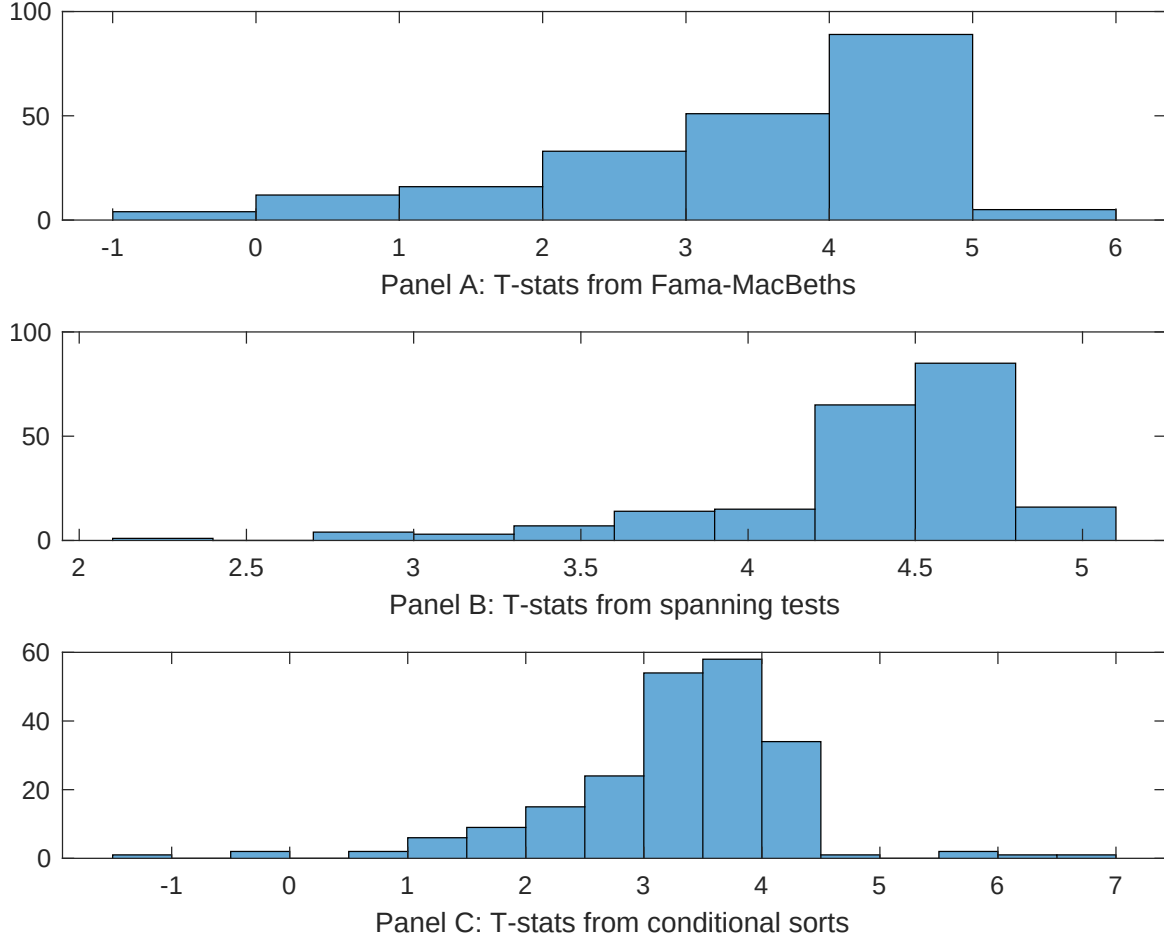


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.99]	0.13 [6.41]	0.17 [6.78]	0.12 [5.42]	0.12 [5.65]	0.13 [6.11]	0.11 [3.67]
EDC	0.47 [3.53]	0.44 [3.12]	0.51 [3.93]	0.46 [3.39]	0.53 [4.10]	0.46 [3.60]	0.31 [2.34]
Anomaly 1	0.14 [5.66]						0.73 [2.53]
Anomaly 2		0.27 [4.89]					0.23 [4.23]
Anomaly 3			0.39 [3.08]				-0.22 [-0.98]
Anomaly 4				0.20 [1.69]			0.98 [0.90]
Anomaly 5					0.12 [3.08]		0.67 [0.11]
Anomaly 6						0.92 [7.49]	0.64 [5.39]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.63]	0.15 [1.93]	0.20 [2.67]	0.24 [3.17]	0.21 [2.78]	0.21 [2.71]	0.15 [2.12]
Anomaly 1	27.66 [7.29]						13.31 [2.91]
Anomaly 2		27.18 [6.75]					12.79 [2.82]
Anomaly 3			32.00 [7.95]				38.42 [6.60]
Anomaly 4				15.70 [5.32]			4.21 [1.32]
Anomaly 5					12.84 [3.36]		-11.62 [-2.17]
Anomaly 6						-0.25 [-0.05]	-16.29 [-3.25]
mkt	2.57 [1.44]	3.12 [1.73]	2.95 [1.66]	4.33 [2.32]	1.66 [0.91]	1.90 [1.03]	4.87 [2.73]
smb	7.70 [2.98]	8.42 [3.23]	5.54 [2.16]	9.14 [3.42]	6.45 [2.43]	6.64 [2.46]	8.94 [3.48]
hml	-8.11 [-2.35]	-5.19 [-1.49]	-11.32 [-3.27]	-13.45 [-3.63]	-9.07 [-2.53]	-7.32 [-2.05]	-11.11 [-3.14]
rmw	-6.65 [-1.64]	-1.41 [-0.37]	9.81 [2.83]	-0.81 [-0.21]	9.47 [2.64]	8.30 [2.30]	-4.98 [-1.18]
cma	16.12 [2.90]	20.11 [3.70]	2.33 [0.37]	20.05 [3.52]	20.40 [3.15]	33.63 [4.40]	9.59 [1.25]
umd	1.84 [1.04]	1.33 [0.74]	2.54 [1.45]	4.36 [2.41]	3.43 [1.88]	3.14 [1.71]	0.53 [0.30]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	13	12	14	10	8	7	21

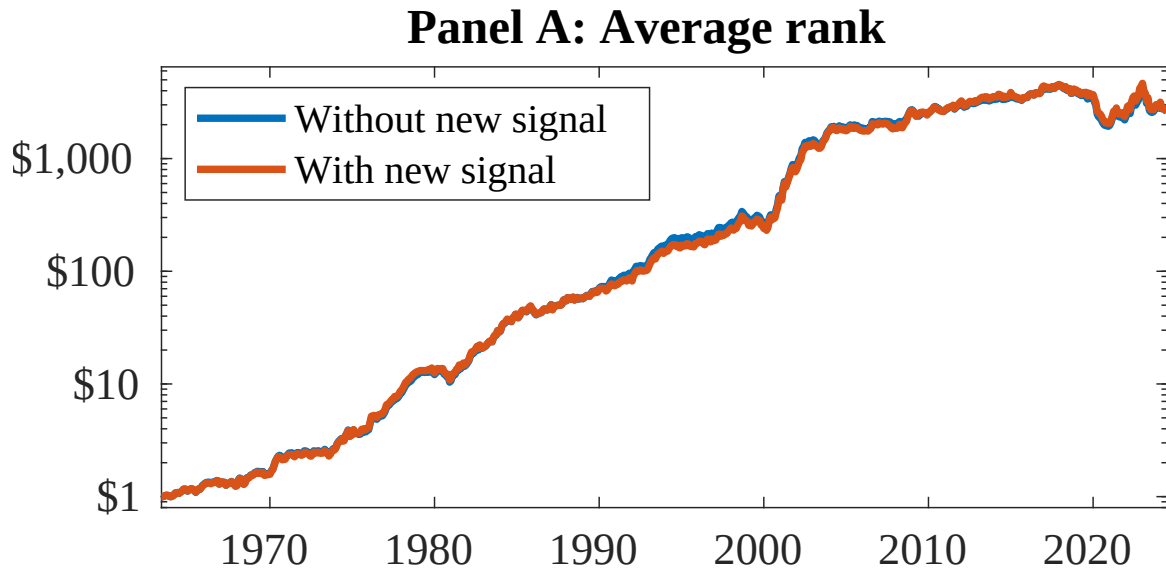


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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